Project Phase 2: Progress Report

Urban Crime Modelling and Prediction in Seattle

A Machine Learning Approach Using Census Tract-Level Socioeconomic and Urban Features

Prince Newman | Hayat Kimo

GitHub Repository:

https://github.com/princenewman02/Urban-Crime-Machine-Learning-Prediction-Seattle

1. Problem Statement

1.1 Background and Motivation

Urban crime remains one of the most complex and socially costly challenges facing cities today.Beyond understanding where crime occurs, we are curious about why certain neighborhoods experience disproportionately high rates relative to others. Traditional criminological approaches have relied on aggregate statistics and reactive policing strategies; however, advances in machine learning and geospatial analysis offer the opportunity to build predictive models that identify the structural and socioeconomic drivers of crime at fine geographic scales.

Seattle, Washington provides an ideal study context. As a rapidly growing tech industry city, Seattle exhibits significant socioeconomic heterogeneity across its 177 census tracts ranging from high-income tech adjacent neighborhoods in North Seattle to lower-income communities in South Seattle alongside rich open-data infrastructure including geocoded police incident records, census tract level American Community Survey data, land use zoning classifications, transit stop locations, and Emergency Medical Services call records.

1.2 Research Questions

This project addresses the following primary research questions:

- What socioeconomic, demographic, and urban structural factors are most strongly associated with serious crime rates across Seattle census tracts?
- Can a machine learning model accurately predict tract-level serious crime rates from observable predictor variables?
- Does the spatial distribution of crime exhibit significant spatial autocorrelation, and if so, does accounting for spatial dependence improve model performance?
- Which census tracts are identified as statistically significant crime hotspots (High-High LISA clusters), and what predictor profiles characterize these tracts?

1.3 Updated Problem Statement

Initial EDA has refined our understanding of the problem in several important ways. First, the dependent variable serious crime rate per 1,000 residents is strongly right-skewed (skewness = 3.70), driven by a small cluster of Downtown and SoDo tracts with extreme crime rates (up to 378 per 1,000). This confirms the need for log-transformation of the dependent variable in regression modeling. Second, police incident rate ($r = 0.98$) and EMS call rate ($r = 0.74$) emerged as dominant predictors, both far exceeding the contribution of traditional socioeconomic predictors. Third, Global Moran's I = 0.216 ($p = 0.001$) confirms significant spatial autocorrelation in crime distribution, indicating that OLS regression alone is insufficient and spatial regression models must be evaluated. Fourth, the bivariate income-crime map revealed that the income-crime relationship is spatially heterogeneous some high-income tracts have high crime due to commercial activity, while the expected low-income/high-crime pattern is concentrated in South Seattle suggesting that spatial interaction terms or geographically weighted regression may improve model fit.

Revised primary question: Which combination of socioeconomic, urban structural, and activity-based predictors best explains tract-level serious crime rates in Seattle, and does accounting for spatial autocorrelation improve predictive accuracy beyond standard OLS regression?

2. Data Preprocessing

2.1 Data Sources

Eight datasets were collected and integrated to construct the master dataset of 177 Seattle census tracts:

Dataset	Source / Description
SPD Crime Data 2022	SPD Crime Data
ACS 5-Year 2022 (B19013)	Median Income
ACS 5-Year 2022 (B23025)	Unemployment Rate
OFM SAEP Block Estimates	Population Density
Current Land Use Zoning	Land Use
King County Transit Stops	Transit Access
Seattle EMS Calls 2022	EMS Calls
Census Tract Boundaries	US Census TIGER/Line 2022 shapefiles — EPSG:4326 polygon boundaries

2.2 Data Cleaning

2.2.1 Crime Data (SPD)

The raw SPD dataset contained 87,733 records. The following cleaning steps were applied:

- REDACTED coordinates removed: 11,604 rows (13.2%) were excluded due to SPD privacy suppression for sensitive offense categories. This is a systemic limitation of the dataset: redacted records are not randomly distributed and may bias crime rate estimates for certain tract types.
- Year filter applied: Only 2022 Offense Start DateTime records were retained (74,760 rows); 1,369 off-year incidents removed.
- Deduplication: SPD assigns multiple offense type rows per Report Number for multi-crime incidents. After deduplication on Report Number, 65,137 unique incidents remained (9,623 duplicates removed).
- Serious crime filter: For the dependent variable, only “Property Crime”and “Violent Crime” offense group categories were retained (43,261 incidents, 67.1% of all incidents). All offense types were retained for the police incident rate predictor.
- Spatial filter: 623 incidents fell outside the Seattle boundary after coordinate projection to EPSG:2926 and spatial joining excluded from tract-level aggregation.

2.2.2 Income Data

The ACS B19013 table was delivered in wide format (1 row, 991 columns). A custom GEOID construction function was developed to correctly convert ACS tract codes to Census GEOID format. The critical fix: whole-number tract codes (e.g., tract "3") must be multiplied by 100 before zero-padding (producing "000300"), not directly padded (which would incorrectly produce "000003"). This resolved a matching failure that initially yielded only 108/177 Seattle tracts. After the fix, 177/177 tracts matched. One suppressed value was set to NaN; no imputation was required as the affected tract was outside Seattle.

2.2.3 Unemployment Data

The same GEOID construction function was applied. One tract with zero labor force (a non-Seattle tract) was set to NaN before computing the unemployment rate. After Seattle filtering, 177/177 tracts matched with zero missing values.

2.2.4 Population Data

The OFM block-level population dataset required cleaning and validation before it could be aggregated to the census-tract level for crime-rate analysis. A full missing-value check confirmed that all population and housing fields including the 2022 variables used in this project contained no nulls, and zeros reflected legitimate uninhabited or water-only blocks. After removing 251 water-only blocks, the remaining 10,042 land blocks showed no missing values or duplicate GEOID_20 entries.

All relevant fields were already stored in appropriate formats, and GEOIDs were preserved as strings to maintain leading zeros. Outlier checks showed strong right-skew in both population and land-area distributions, but these extreme values represented real urban variation and were retained. Validation steps confirmed that selecting only GEOID_20, Population 2022, and Land Acres produced the correct variables for aggregation, and grouping by TRACT_GEOID yielded 177 Seattle census tracts with no missing or zero land-area values. The final tract-level dataset includes GEOID, Population 2022, Land Acres, and population density, with complete values for all 177 tracts, providing a reliable denominator for crime-rate calculations and neighborhood-level modeling.

2.2.5 Seattle EMS Calls 2022

The EMS dataset was already clean and fully within 2022, with all 10,842 records confirmed as Medic Response calls. After removing 330 rows with missing or invalid coordinates, 10,812 incidents were successfully geocoded and converted to point geometry. Reprojecting to EPSG:2926 and performing a spatial join assigned 10,772 calls to Seattle tracts, with only 40 falling outside the boundary. Every one of the 177 tracts received at least one EMS call, and the resulting call-rate distribution was strongly right-skewed due to high-activity downtown tracts. The final tract-level dataset contains 177 rows and four complete fields GEOID, ems_call_count, population, and EMS_call_rate and is ready for integration into the master dataset.

2.2.6 Transit Stops

The GTFS stops dataset contained 6,351 stops, all with valid latitude and longitude values and no missing or zero coordinates, confirming the file was fully complete prior to processing. All stops were converted to point geometry in EPSG:4326 and successfully reprojected to EPSG:2926 for spatial alignment with Seattle census tracts. The spatial join reduced the regional dataset to 2,594 stops within Seattle, with the remaining 3,757 stops (59.2%) located in surrounding King County cities. These 2,594 stops were distributed across 175 of 177 tracts, leaving two tracts with zero stops, which were assigned a transit access value of 0. Stop counts ranged from 1 to 79 per tract, showing moderate right skew, and were converted into a density-adjusted transit access measure ranging from 2.65 to 212.80 stops per square mile. This right-skewed distribution reflects Seattle's uneven transit coverage, with extremely high densities in compact downtown tracts. The resulting transit_access variable is clean, complete, and ready for integration into the master dataset.

2.2.7 Land Use Zoning

The zoning data was cleaned by reprojecting all parcels to EPSG:2926 and repairing all 17 invalid geometries, resulting in a fully valid set of 3,619 polygons ready for spatial analysis. Commercial parcels were identified using the four confirmed zoning categories, yielding 1,391 commercial parcels (38.4%). A spatial join assigned 3,592 parcels to Seattle tracts, with only 27 parcels falling outside tract boundaries. Parcel areas were computed and aggregated to the tract level, producing valid total-area and commercial-area values for 175 tracts, with the remaining two tracts receiving a commercial ratio of 0. The final commercial land-use ratio ranges from 0 to 1 and accurately reflects Seattle's mixed-use patterns, producing a clean 175-row dataset ready for integration into the master dataset.

2.2. General Cleaning Standards

- Currency strings: Commas removed from income values; "250,000+" top-coded values converted to 250,000.
- Suppressed values: Census suppression codes ("-", "N", "**") converted to NaN before numeric conversion.
- Zero population tracts: One tract with zero estimated population replaced with NaN to prevent division-by-zero in rate computations.

2.3 Data Integration

All eight datasets were integrated into a single master GeoDataFrame using sequential left merges on GEOID. The merge was consolidated into a single cell to prevent duplicate column suffix conflicts (_x/_y) from multiple reruns. Final integration result:

Dimension	Value	Notes
Rows	177	All Seattle census tracts
Columns	11	8 variables + GEOID + NAME + geometry
Missing Values (pre-impute)	2	commercial_land_use_ratio only
Missing Values (post-impute)	0	Imputed with Seattle median (0.18)
GEOID duplicates	0	All GEOIDs unique

The two missing commercial land use ratio values arose from census tracts where no zoning parcels intersected the tract boundary in the spatial join; these were imputed with the Seattle citywide median commercial ratio (0.18).

2.4 Data Normalization and Feature Engineering

- Rate normalization: Crime, EMS, and police counts were converted to per-1,000-resident rates by dividing by Office of Financial Management (OFM) population estimates and multiplying by 1,000. Transit stops were normalized to per-square-mile by dividing by tract area in square miles.
- Commercial land use ratio: Computed as the proportion of total parcel area classified as commercial zoning within each census tract (range 0.0 to 1.0).

- Log transformation: Six variables exhibit skewness above the threshold of 1.0 and will be log-transformed (using log1p) for Phase 5 regression modeling: crime_rate (3.70), population_density (3.09), transit_access (2.21), unemployment_rate (1.65), police_incident_rate (4.56), EMS_call_rate (5.28). Median income (0.09) and commercial_land_use_ratio (0.96) are sufficiently symmetric and will be used on their original scale.
- Coordinate systems: All spatial data was projected to EPSG:2926 (Washington State Plane South, feet) for spatial operations. The final GeoJSON export was reprojected to EPSG:4326 (WGS84) for standard GeoJSON compatibility.

3. Preliminary Results of Exploratory Data Analysis

3.1 Descriptive Statistics

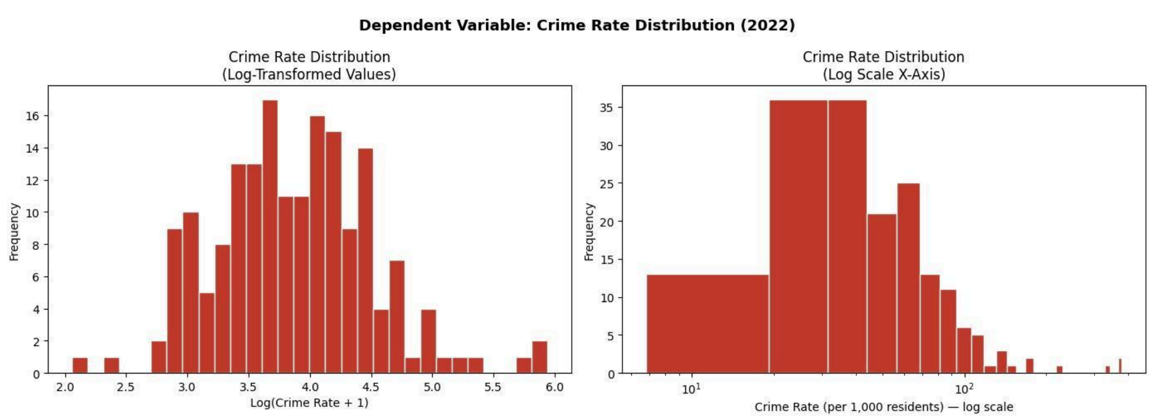
Table 3.1 presents descriptive statistics for all variables in the master dataset (n = 177 tracts):

Variable	Mean	Median	Std	Min	Max	Skewness
Crime Rate (per 1K)	59.08	45.43	52.33	6.81	378.19	3.70
Population Density	25.84	18.13	26.42	1.35	203.42	3.09
Median Income ($K)	120.62	121.18	43.08	19.38	250.00	0.09
Unemployment Rate	4.34%	3.36%	3.11%	0.00%	17.60%	1.65
Commercial Ratio	0.21	0.13	0.24	0.00	1.00	0.96
Transit Access	33.18	24.93	34.04	0.00	212.80	2.21
EMS Call Rate	20.96	14.78	22.73	1.36	146.84	5.28
Police Incident Rate	113.94	69.14	120.55	10.08	766.12	4.56

Table 3.1: Descriptive statistics for all master dataset variables (n = 177 Seattle census tracts, 2022)

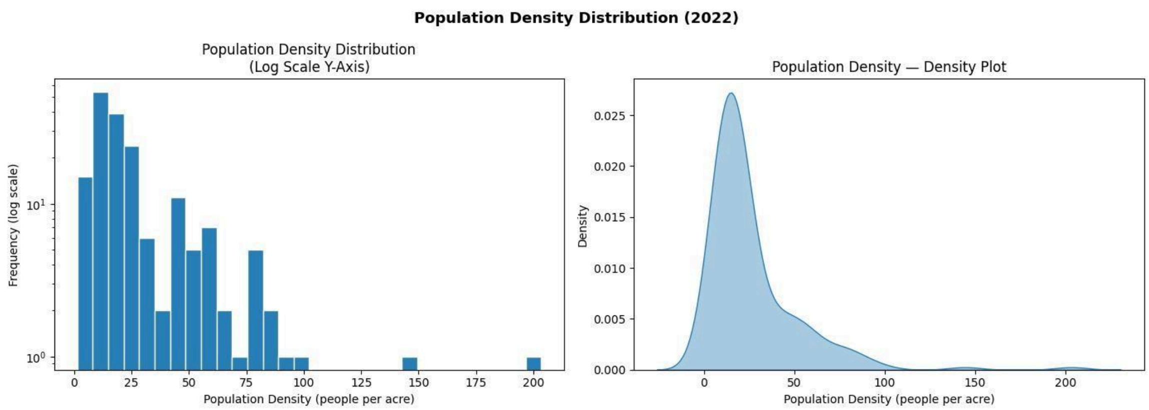
3.2 Univariate Distribution Analysis

Figures 3.1 through 3.8 present the distribution of each variable. Visualization types were selected to match the distributional characteristics of each variable: log-transformed histograms for heavily right-skewed variables, empirical CDFs for near-normal variables, violin plots for moderate skew, and KDE with rug plot for the bimodal commercial ratio.



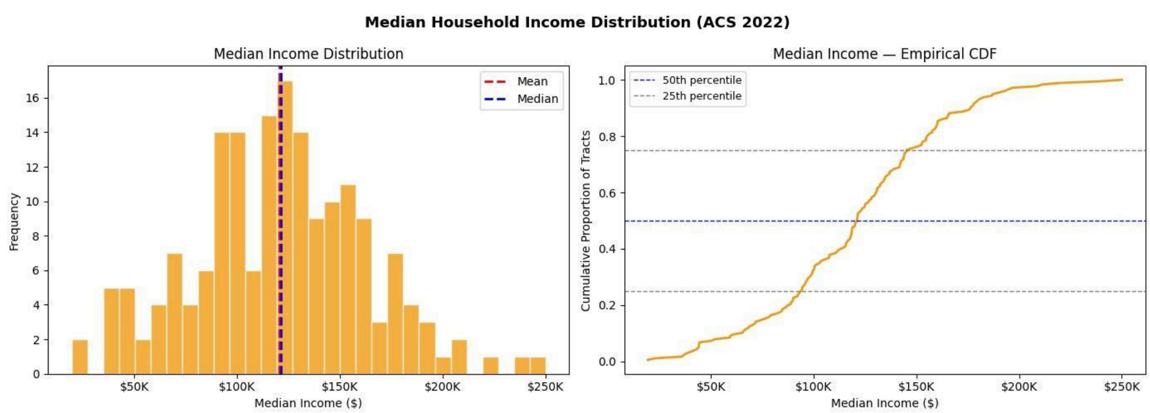
Crime Rate Log-transformed histogram (left) and log scale x-axis (right). Skewness = 3.70. Log transformation reveals near-normal shape.

Plot Observation: Crime rate across Seattle's 177 census tracts is strongly right skewed with an original skewness of 3.70, ranging from 6.81 to 378.19 serious crimes per 1,000 residents with a mean of 59.08 and median of 45.43. The log-transformed histogram reveals a near-normal distribution shape hidden by the skew, confirming that a log transformation of the dependent variable will be evaluated in Phase 5 to meet OLS normality assumptions. The log scale x-axis plot confirms that the majority of tracts fall between 10 and 100 crimes per 1,000 residents with only a small number of extreme outliers above 200, likely Downtown and high-activity commercial tracts whose large daytime populations are not captured in residential census counts.



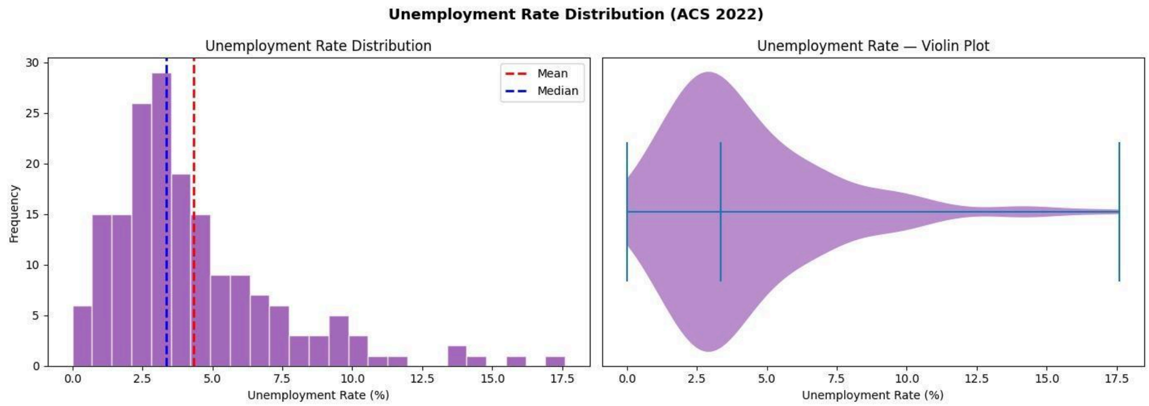
Population Density Log-transformed histogram and KDE. Skewness = 3.09. Majority of tracts between 5–30 people/acre.

Plot Observation: Strongly right-skewed (3.09); the majority of Seattle tracts have densities between 5–30 people per acre, reflecting typical residential neighborhoods, while a small number of high-density urban core tracts exceed 100 people per acre. Log transformation required before modeling.



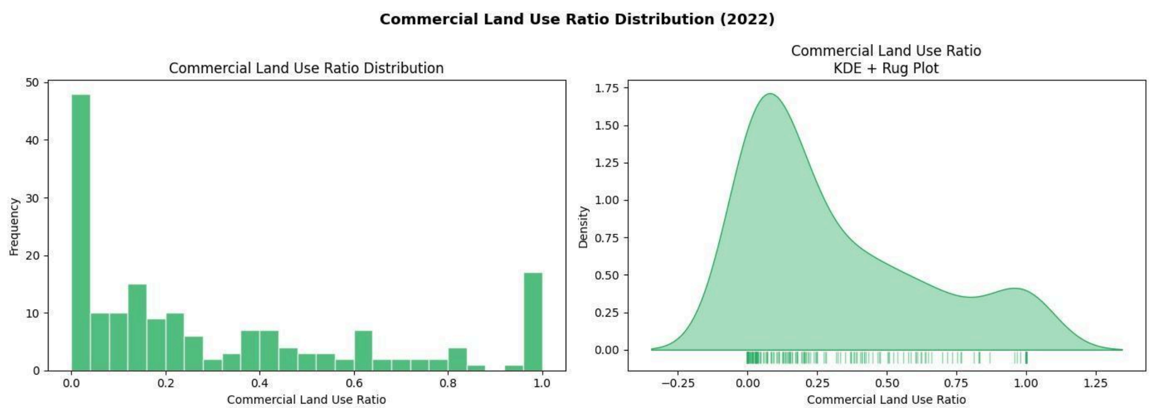
Median Income Histogram with mean/median lines and Empirical CDF. Skewness = 0.09. Near-symmetric, centered at $121K. No transformation needed.

Plot Observation: Strongly right-skewed (3.09); the majority of Seattle tracts have densities between 5–30 people per acre, reflecting typical residential neighborhoods, while a small number of high-density urban core tracts exceed 100 people per acre. Log transformation required before modeling.



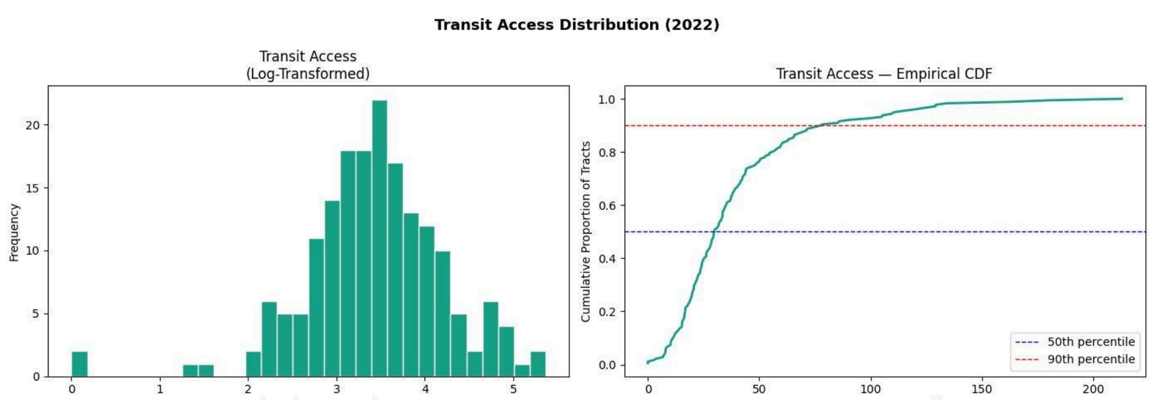
Unemployment Rate — Histogram and violin plot. Skewness = 1.65. Most tracts between 2–5%. Long right tail to 17.5%.

Plot Observation: Right-skewed (1.65) with most tracts clustered between 2–5%, the histogram and violin plot both confirm the bulk of Seattle tracts have low unemployment with a long right tail driven by a small number of high-unemployment tracts reaching 17.5%. The mean (4.34%) sits visibly to the right of the median (3.36%) confirming the skew.



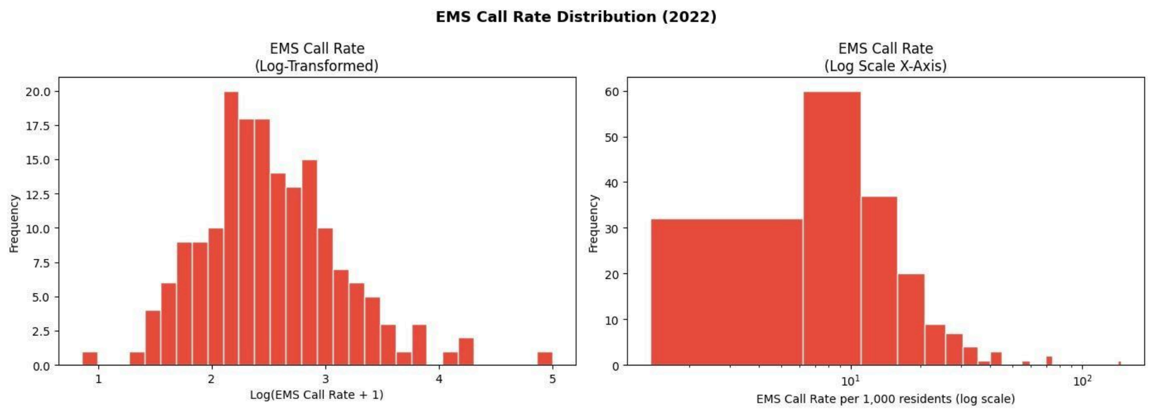
Commercial Land Use Ratio Histogram and KDE+rug plot. Bimodal distribution with spikes at 0 (48 purely residential tracts) and 1.0 (17 entirely commercial tracts).

Plot Observation: Bimodal distribution with two distinct spikes, a dominant spike at 0 (48 purely residential tracts with no commercial land) and a secondary spike at 1.0 (17 entirely commercial tracts). The KDE + rug plot confirms this U-shaped pattern clearly. Most tracts fall below 0.25 commercial ratio with sparse coverage across the middle range.



Transit Access Log-transformed histogram and Empirical CDF. Skewness = 2.21. 90th percentile at ~100 stops/sq mile; most tracts below 50

Plot Observation: Right-skewed (2.21) but log transformation reveals a near-normal distribution. The ECDF shows 50% of tracts have fewer than ~30 stops per square mile and 90% have fewer than ~100, confirming that high-density transit coverage is concentrated in a small number of urban core tracts while most Seattle tracts have moderate access.



EMS Call Rate Log-transformed histogram and log scale x-axis. Skewness = 5.28 most extreme skew in the dataset.

Plot Observation: Right-skewed (2.21) but log transformation reveals a near-normal distribution. The ECDF shows 50% of tracts have fewer than ~30 stops per square mile and 90% have fewer than ~100, confirming that high-density transit coverage is concentrated in a small number of urban core tracts while most Seattle tracts have moderate access.

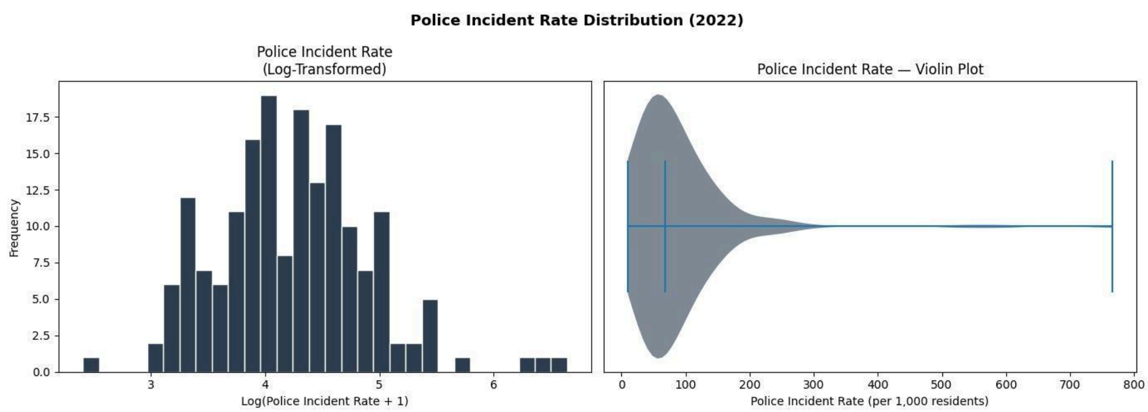
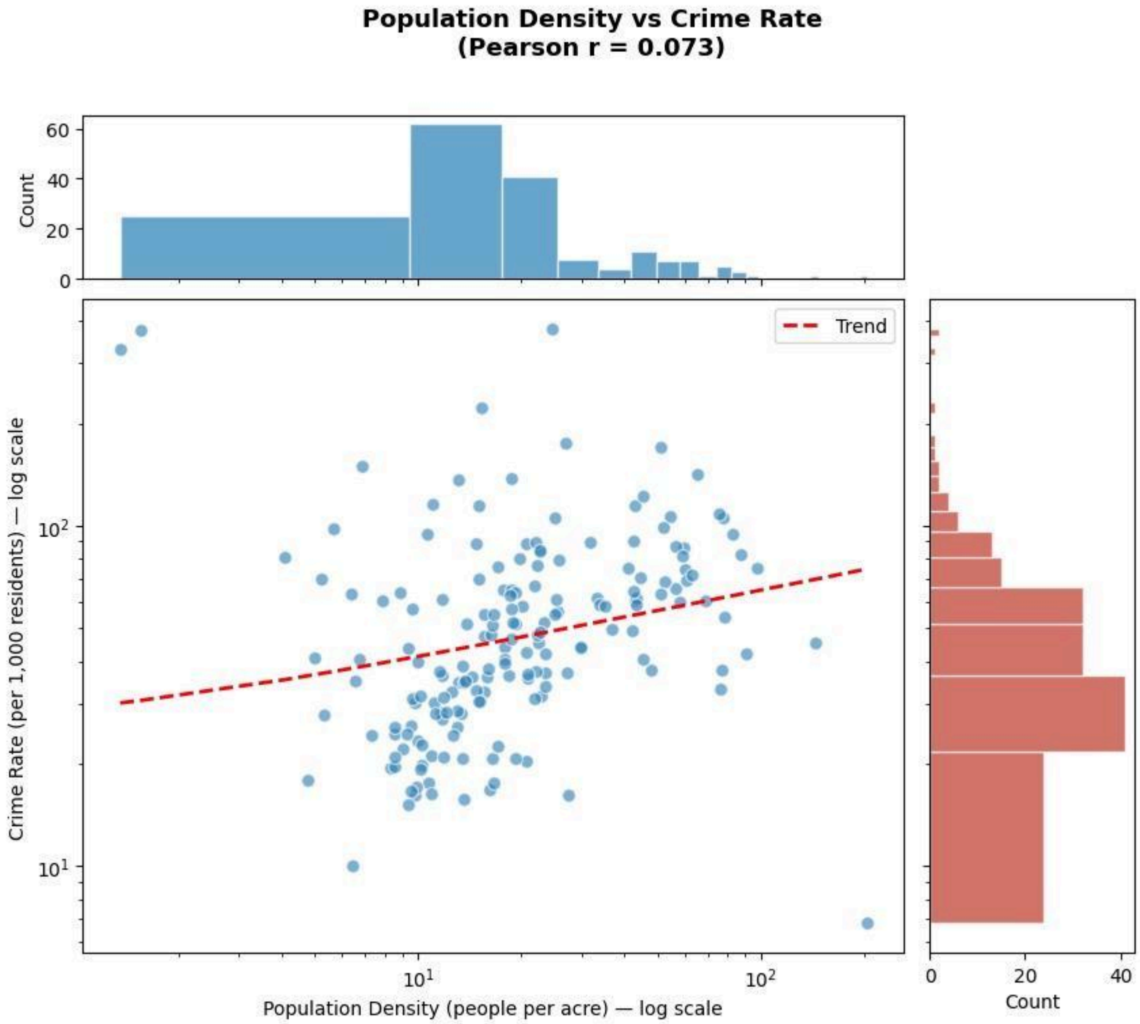


Figure 3.8: Police Incident Rate Log-transformed histogram and violin plot. Skewness = 4.56. Extreme outlier at 766/1,000 residents (Downtown commercial tract).

Plot Observation: Extremely right-skewed (4.56), the log-transformed histogram shows a broad but roughly normal shape centered around log(4)–log(5), corresponding to roughly 50–150 incidents per 1,000 residents. The violin plot dramatically illustrates the extreme outliers, the long thin tail extending to 766 incidents per 1,000 residents is entirely driven by 2–3 Downtown tracts with disproportionately high activity relative to their residential populations. Log transformation essential.

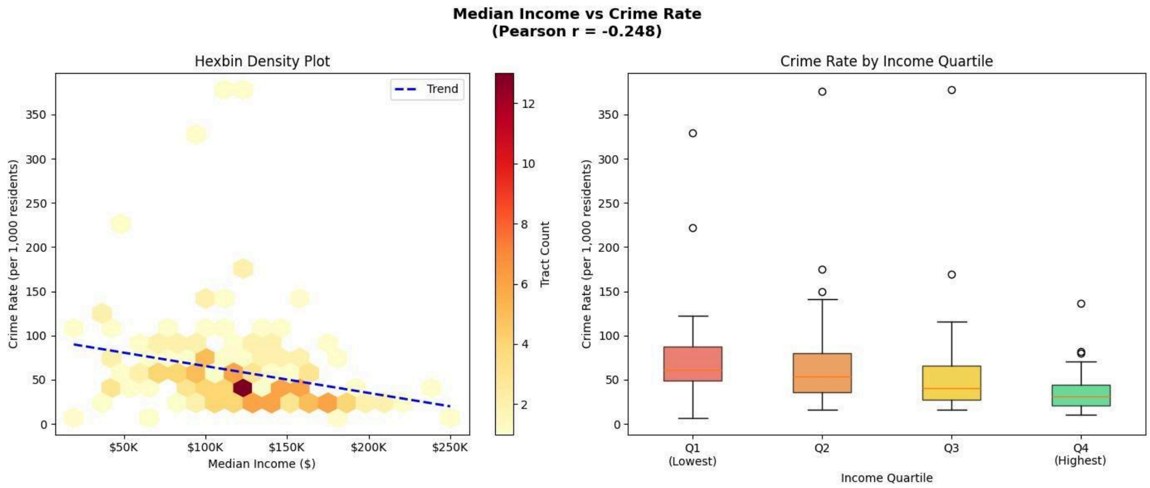
3.3 Bivariate Analysis - Predictors vs Crime Rate

Figures 3.9 through 3.15 present bivariate relationships between each predictor and crime rate. Visualization types were varied to reveal different aspects of each relationship: log-log scatter with marginal histograms, hexbin density plots, quintile bar charts, categorical boxplots, binned mean line charts, and 2D KDE contour plots.



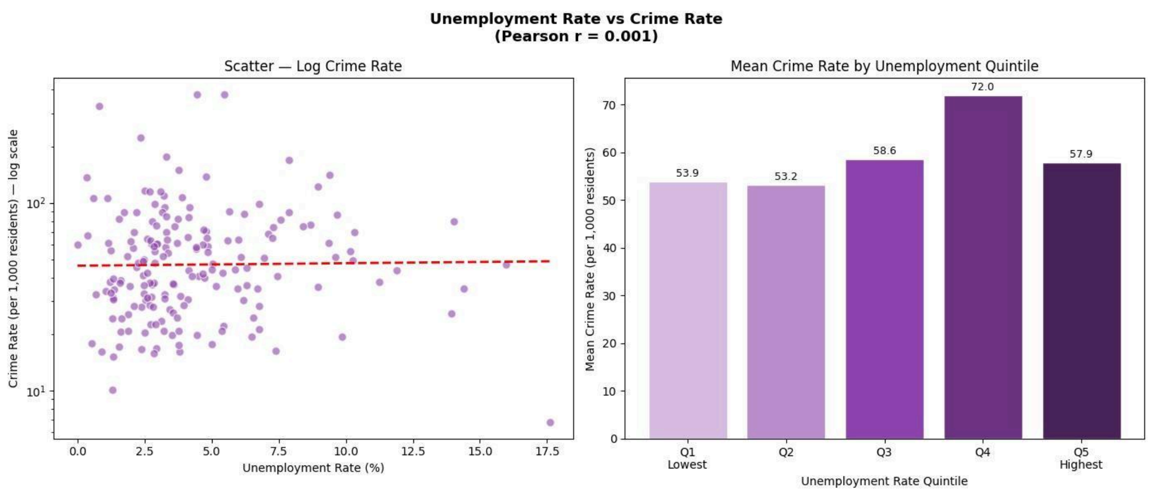
Population Density vs Crime Rate (r = 0.073). Log-log scatter with marginal distributions. Near-zero correlation dense tracts are not systematically more crime-prone.

Plot Observation: Near-zero correlation; population density has virtually no linear relationship with crime rate. The log-log scatter shows wide vertical spread at every density level, confirming that densely populated tracts are not systematically more or less crime-prone than sparse ones.



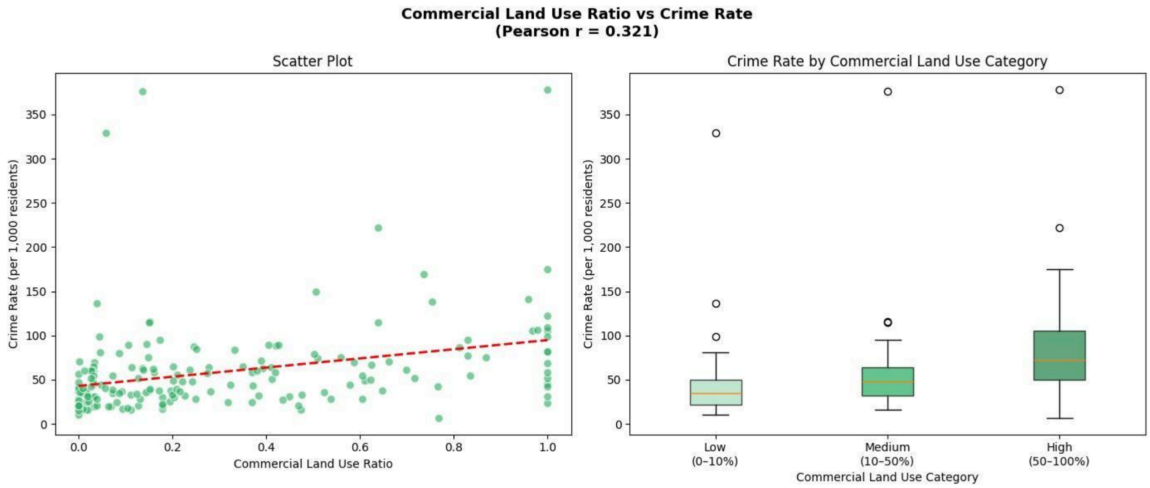
Median Income vs Crime Rate (r = -0.248). Hexbin density plot + quartile boxplots. Weak negative relationship; median crime declines from Q1 ($70) to Q4 ($30) but with wide IQRs.

Plot Observation: Weak negative relationship, higher income tracts tend toward lower crime rates but with substantial overlap. The quartile boxplot clearly shows a stepwise decline in median crime from Q1 (70 median) to Q4 (30 median), though wide IQRs and outliers in every quartile indicate income alone is a weak predictor of crime.



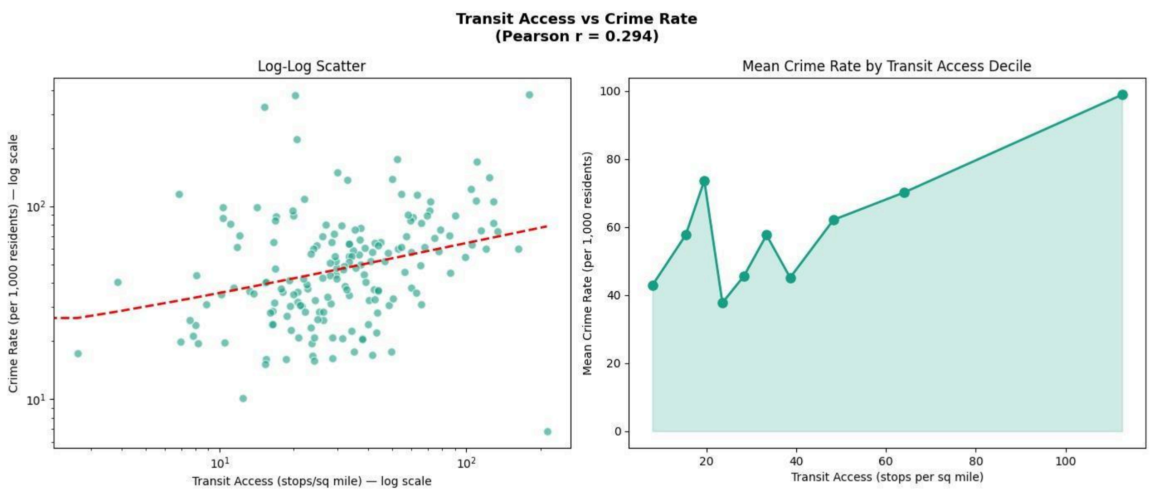
Unemployment Rate vs Crime Rate (r = 0.001). Scatter + quintile bar chart. Essentially zero linear correlation. Trend line is completely flat.

Plot Observation: Essentially zero linear correlation the scatter shows no directional trend and the trend line is completely flat. Surprisingly the quintile bar chart shows Q4 (72.0) slightly elevated above the others, suggesting a possible non-linear relationship at very high unemployment levels, but this is not captured by Pearson r.



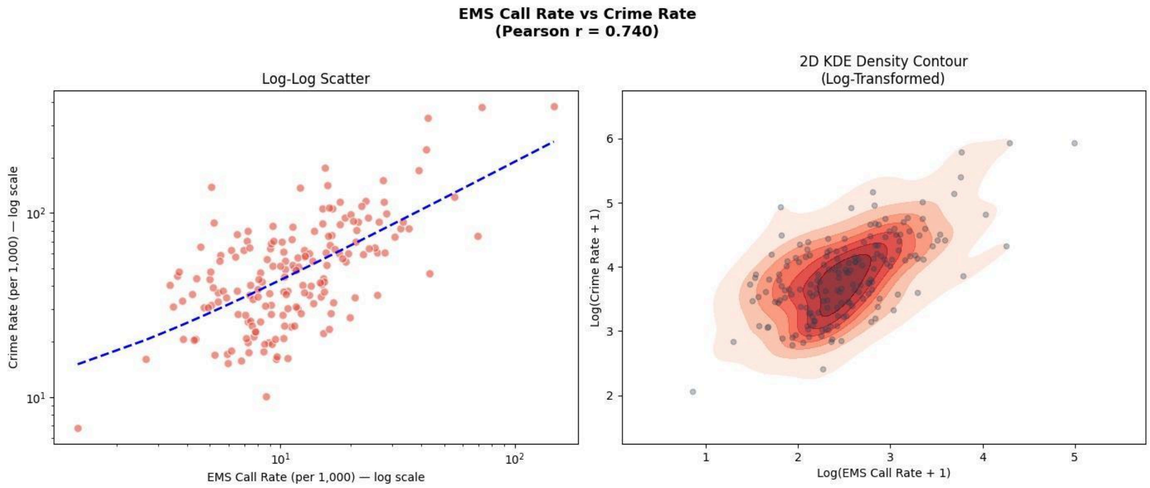
Commercial Land Use Ratio vs Crime Rate (r = 0.321). Scatter + categorical boxplot. Moderate positive relationship median crime rises from ~35 (Low commercial) to ~70 (High commercial).

Plot Observation: Moderate positive relationship tracts with higher commercial land use have higher crime rates. The categorical boxplot confirms this clearly median crime rises from ~35 in Low commercial tracts to ~70 in High commercial tracts. Consistent with routine activity theory where commercial areas attract more offenders and targets.



Transit Access vs Crime Rate (r = 0.294). Log-log scatter + binned mean line chart. Moderate positive trend mean crime rises from ~42 (lowest decile) to ~99 (highest decile).

Plot Observation: Moderate positive relationship tracts with higher commercial land use have higher crime rates. The categorical boxplot confirms this clearly median crime rises from ~35 in Low commercial tracts to ~70 in High commercial tracts. Consistent with routine activity theory where commercial areas attract more offenders and targets.



EMS Call Rate vs Crime Rate (r = 0.740). Log-log scatter + 2D KDE contour. Strong positive relationship tight diagonal density band confirms EMS rate as a reliable crime proxy.

Plot Observation: Moderate positive relationships tract with more transit stops tend toward higher crime rates. The decile line chart shows a generally upward trend with mean crime rising from ~42 in the lowest transit decile to ~99 in the highest, though with some noise in the middle deciles suggesting the relationship is not strictly linear.

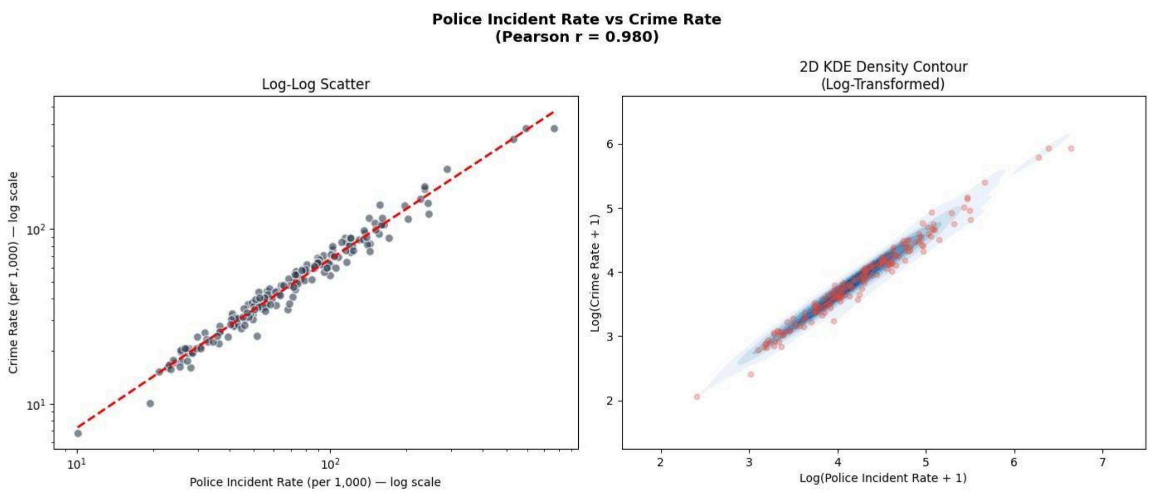


Figure 3.15: Police Incident Rate vs Crime Rate (r = 0.980). Log-log scatter + 2D KDE contour. Near-perfect correlation points align almost exactly on the trend line. Multicollinearity concern flagged.

Plot observation: Near perfect positive correlation is by far the strongest predictor in the dataset. The log-log scatter shows points aligned almost perfectly along the trend line with minimal scatter, and the 2D KDE contour collapses to an extremely narrow diagonal band confirming that police incident rate and crime rate are essentially measuring the same underlying phenomenon. This raises a potential multicollinearity concern that must be carefully evaluated in Phase 5 before including both variables in the same model.

3.4 Correlation Analysis

The Pearson correlation matrix, network graph, and lollipop chart collectively reveal the full structure of predictor relationships.

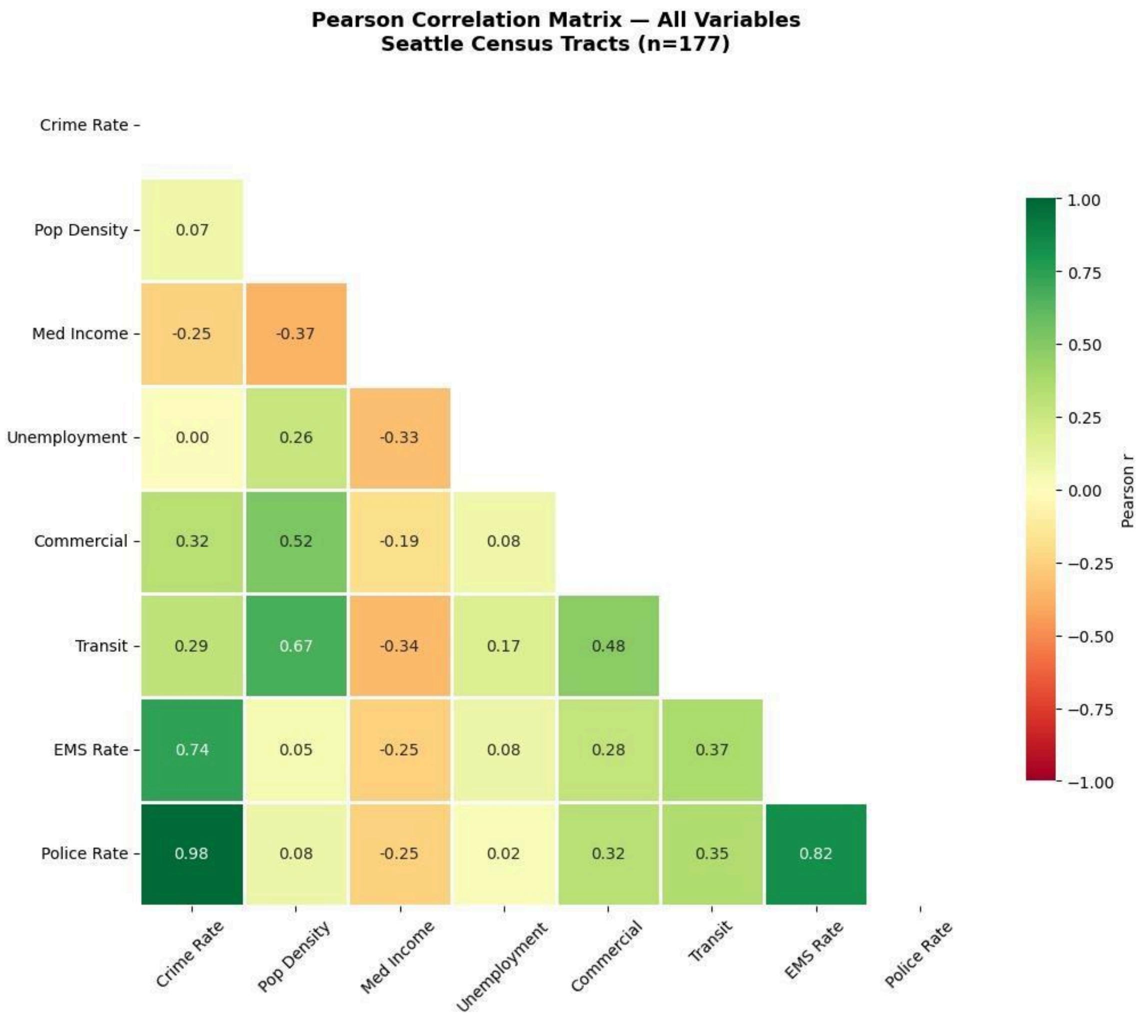
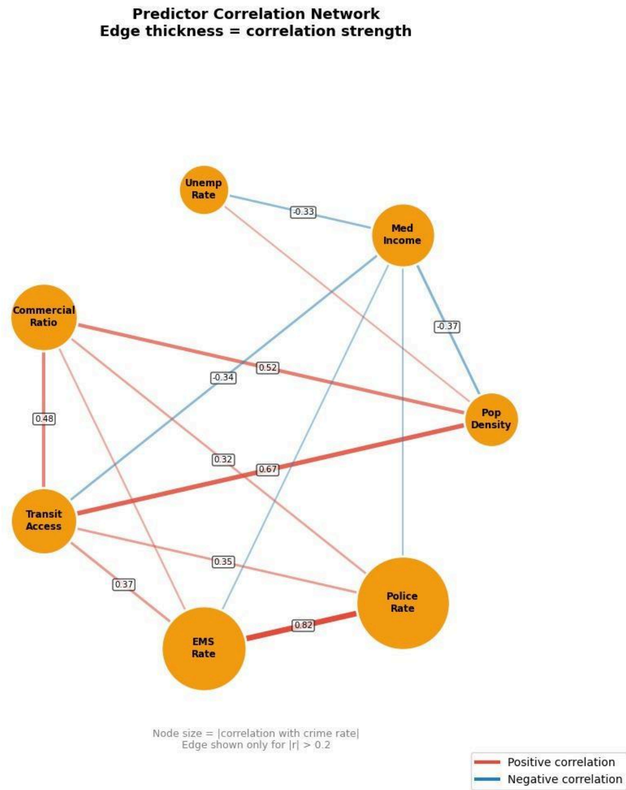


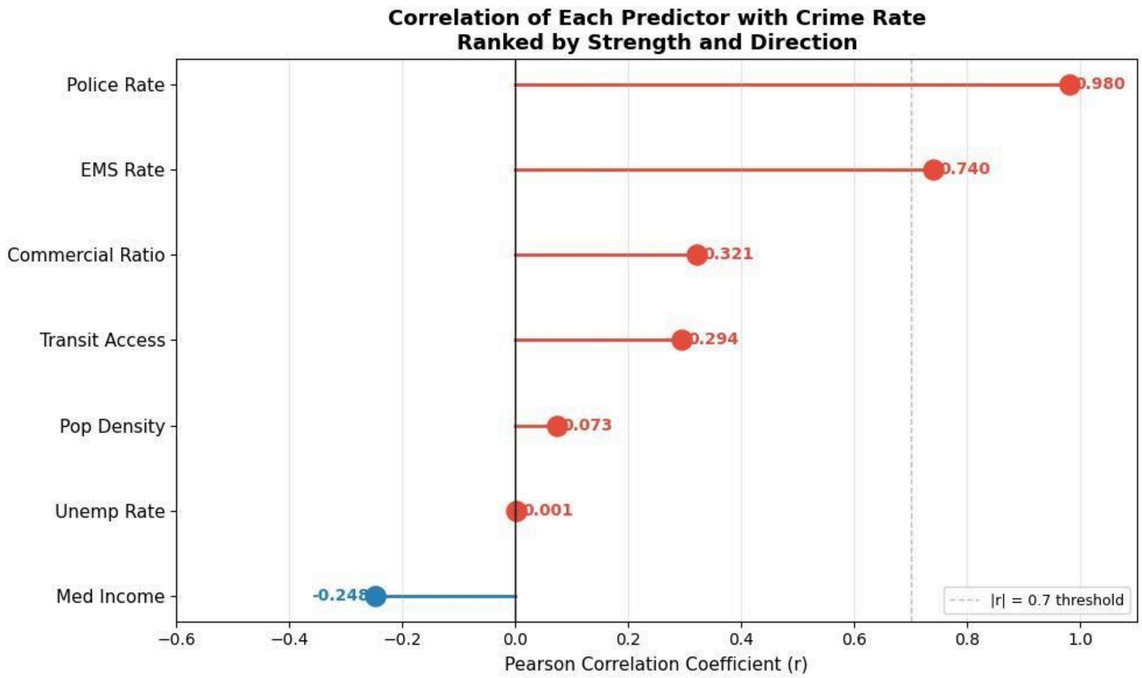
Figure 3.16: Pearson Correlation Heatmap (lower triangle). Two dominant clusters: (1) Police rate, EMS rate, crime rate strongly interconnected. (2) Population density, commercial ratio, transit access moderately correlated urban intensity cluster.

Plot Observation: The correlation matrix reveals two dominant patterns. First, police incident rate (r = 0.98) and EMS call rate (r=0.74) are by far the strongest predictors of crime rate both capturing overall neighborhood activity and disorder. Second, the structural predictors commercial ratio (r = 0.32), transit access (r = 0.29), and median income (r = -0.25) show weak to moderate relationships while population density (r = 0.07) and unemployment (r = 0.001) show virtually no linear relationship with crime rate.



Predictor Correlation Network Graph. Node size proportional to |correlation with crime rate|. Thickest edge: EMS rate vs Police rate (r = 0.822) the only predictor pair exceeding the multicollinearity threshold of |r| > 0.7.

Plot Observation: The network graph clearly isolates EMS rate and police rate as the most crime connected nodes, their large size reflects high absolute correlation with crime rate. The thickest edge in the entire network is EMS vs police (r = 0.82) a potential multicollinearity concern flagged for Phase 5. Population density sits at the center of a web of moderate positive correlations with commercial ratio (0.52), transit (0.67), and police rate (0.32) confirming it proxies urban intensity rather than crime directly. Income is the only node with predominantly blue (negative) edges, sitting structurally opposed to density and transit.



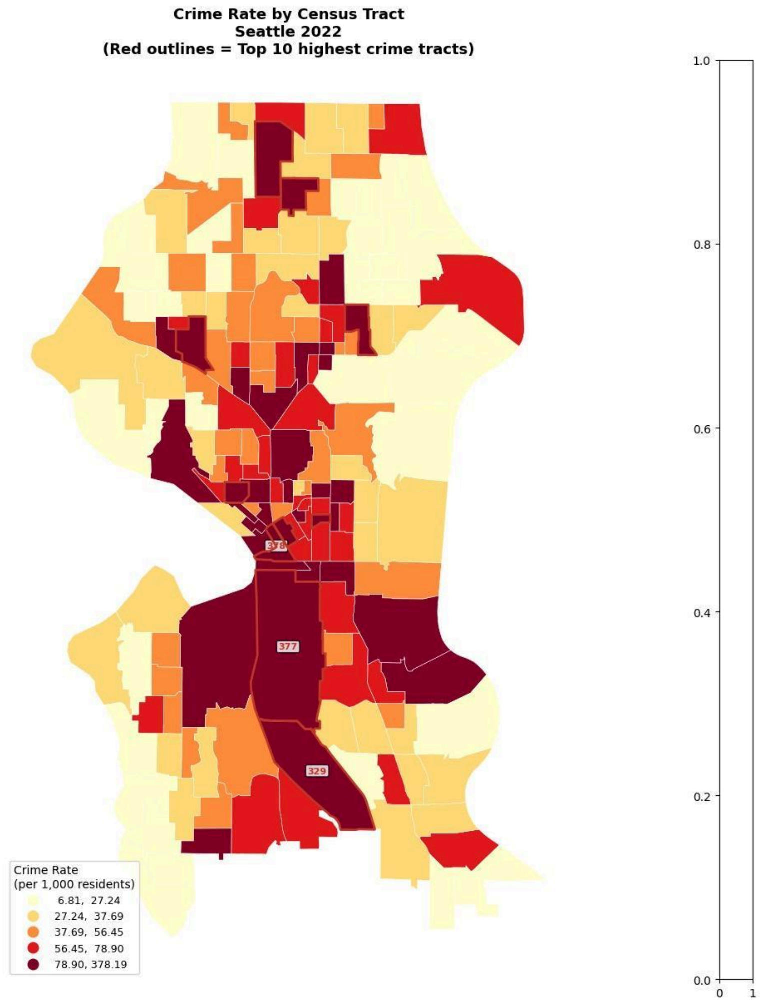
Correlation of Each Predictor with Crime Rate (ranked lollipop chart). Police rate (0.98) and EMS rate (0.74) are in a completely different league from all other predictors. Only median income is negatively correlated (-0.25).

Key correlation findings:

- One highly correlated predictor pair identified: EMS call rate vs police incident rate ($r = 0.822$), exceeding the $|r| > 0.7$ multicollinearity threshold. Both variables will not be included simultaneously in the same OLS model without VIF testing.
- Population density and transit access are moderately correlated ($r = 0.667$) and both correlate with commercial ratio ($r = 0.521$, $r = 0.484$) suggesting an underlying urban intensity factor.
- Median income is negatively correlated with population density ($r = -0.368$) and transit access ($r = -0.339$), reflecting Seattle's spatial structure where denser, transit-rich neighborhoods tend to have lower median incomes.

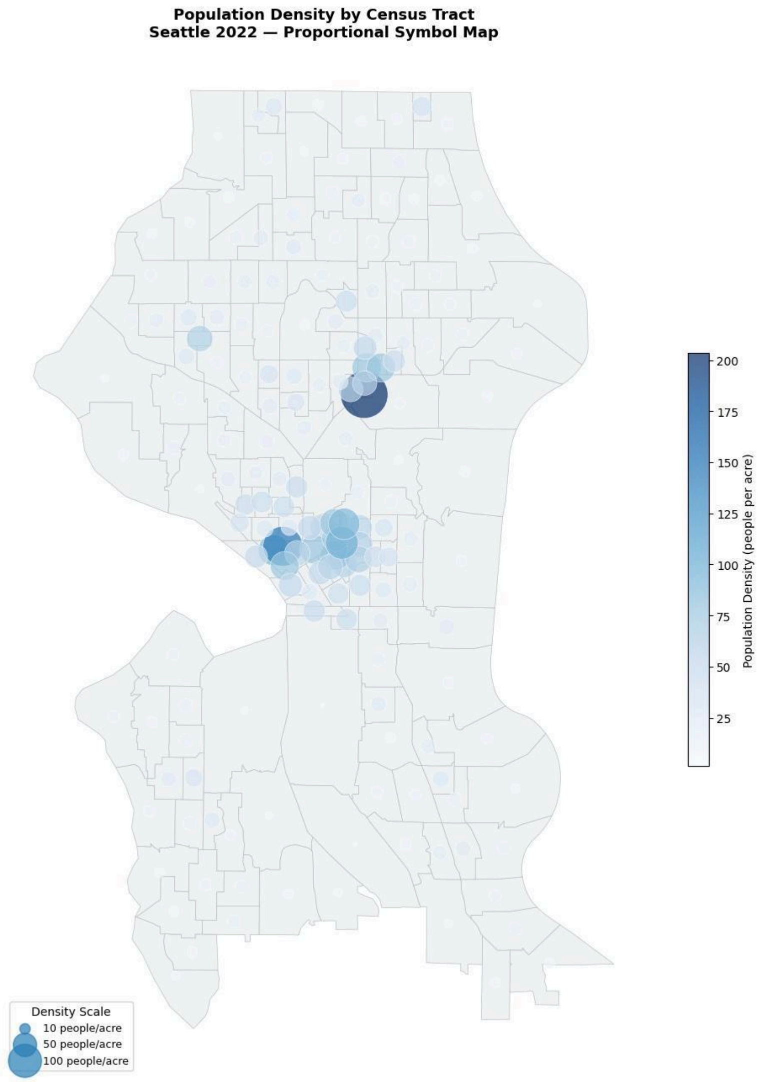
3.5 Spatial Exploratory Analysis

Choropleth maps and spatial visualization reveal the geographic distribution of crime and predictor variables across Seattle's 177 census tracts.



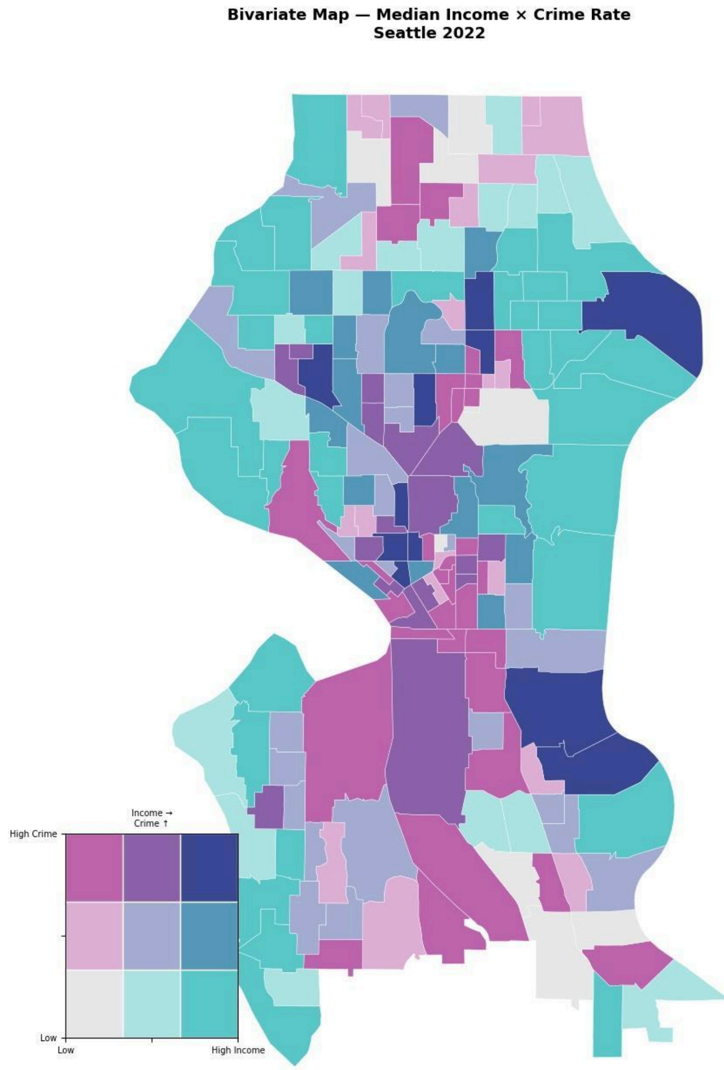
Crime Rate Choropleth (quantile classification, 5 classes). Top 3 crime tracts annotated. Heavy concentration in Downtown/SoDo corridor. Residential periphery consistently low.

Observation: Crime is heavily concentrated in Downtown and South Seattle, the darkest tracts (78–378 per 1,000) form a continuous north-south corridor through the urban core. The three annotated peak tracts (378, 377, 329) are all clustered in the Downtown/SoDo area. Northern and western residential neighborhoods show consistently low rates (6.81–27), confirming a strong center-periphery spatial gradient in crime distribution.



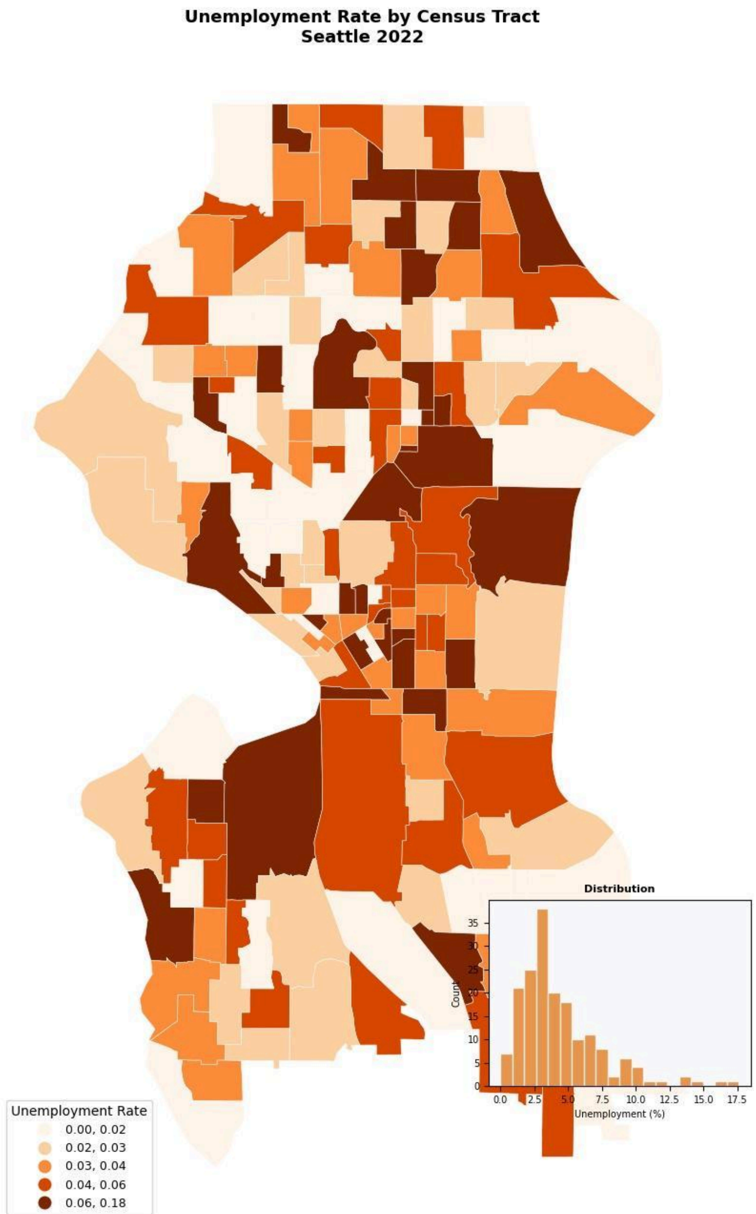
Population Density Proportional Symbol Map. High-density clusters at Capitol Hill/First Hill and University District. South Seattle shows lower residential density.

Observation: High-density tracts are tightly concentrated in two clusters, Capitol Hill/First Hill in the center and the University District in the north with the largest darkest circles confirming extreme density (100–200+ people per acre) in these areas. The rest of Seattle is characterized by small pale circles reflecting low to moderate residential density, and the largely empty south confirms lower-density land use patterns there.



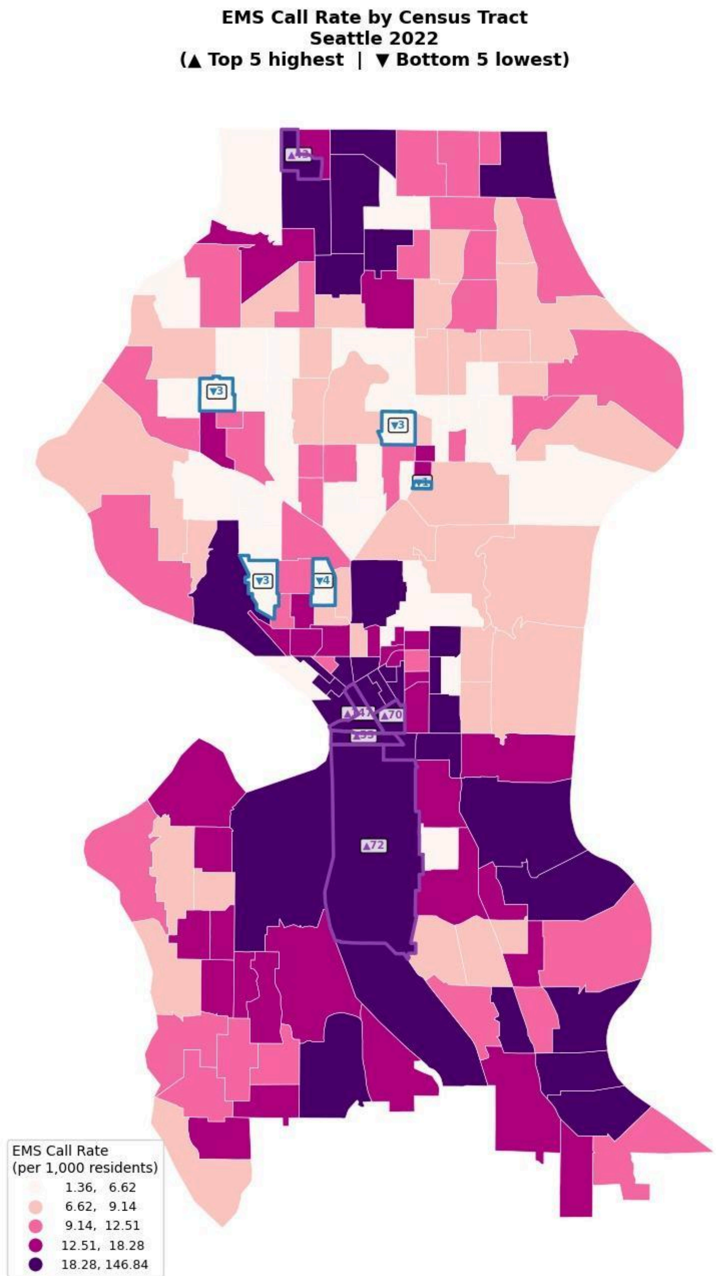
Bivariate Map Median Income x Crime Rate (3x3 color matrix). Dark blue = high income/high crime (Downtown). Teal = high income/low crime (North Seattle). Pink/magenta = low income/high crime (South Seattle). Reveals spatial heterogeneity missed by Pearson r = -0.25.

Observation: The bivariate map reveals a nuanced spatial story invisible in a simple correlation. Dark blue tracts (high income, high crime) appear in Downtown and Capitol Hill, wealthy neighborhoods with high crime driven by commercial activity rather than poverty. Teal tracts (high income, low crime) dominate the northern and eastern residential neighborhoods. Pink/magenta tracts (low income, high crime) cluster in South Seattle, the pattern most aligned with traditional crime-poverty theory. The presence of multiple color types confirms why the simple Pearson r of -0.25 misses the spatial complexity of the income-crime relationship.



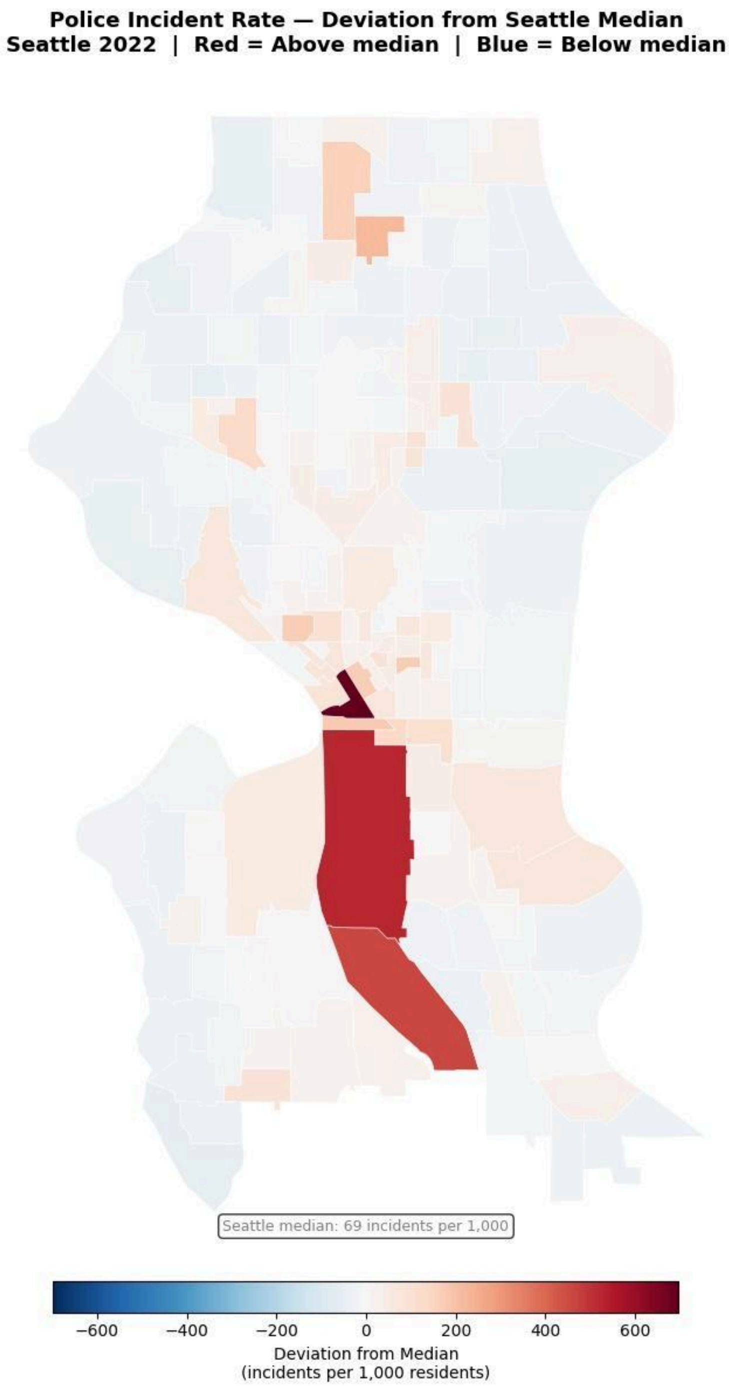
Unemployment Rate Choropleth with inset histogram. Fragmented spatial pattern high unemployment tracts scattered across the city rather than forming a contiguous zone. Consistent with r = 0.001.

Observation: cluster high unemployment tracts (dark brown, 6-18%) are scattered across North Seattle, the Central District, and pockets of South Seattle rather than forming a contiguous zone. The inset histogram confirms the right-skewed distribution with most tracts below 5% the dark tracts visible on the map represent the long tail of the distribution. This fragmented pattern is consistent with the near zero correlation with crime rate (r=0.001).



EMS Call Rate Choropleth with top/bottom 5 tract annotations. Top 5 tracts (47–99 per 1,000) all within Downtown core. Bottom 5 (3–4 per 1,000) in quieter residential areas. Spatial footprint mirrors crime rate map.

Observation: EMS demand is heavily concentrated in Downtown, SoDo, and South Seattle; the deep purple zone in the lower center mirrors the crime rate hotspot almost exactly, consistent with the strong r = 0.74 correlation. The top 5 highest tracts (47, 70, 99, 72, 42) are all within or immediately adjacent to the Downtown core. The bottom 5 lowest tracts (3–4 per 1,000) are scattered across quieter residential areas in the north and west confirming EMS demand as a reliable spatial proxy for neighborhood distress.



Police Incident Rate Diverging Deviation Map (centered on Seattle median of 69 per 1,000). Virtually the entire city sits at or below the median (light blue/white). Extreme positive deviations (deep red, 200–700+ above median) confined to Downtown/SoDo a small cluster of tracts driving the r = 0.98 correlation.

Plot Observation: The diverging map makes the extreme concentration of police activity unmistakably clear virtually the entire city sits at or below the Seattle median of 69 incidents per 1,000 (light blue to white) while a small cluster of Downtown and SoDo tracts show extreme positive deviation of 200–700+ incidents above the median (deep red). This extreme concentration in just a handful of tracts explains the high standard deviation (88.35) and is the spatial signature of the r=0.98 correlation with crime rate; both variables are measuring the same Downtown concentrated phenomenon.

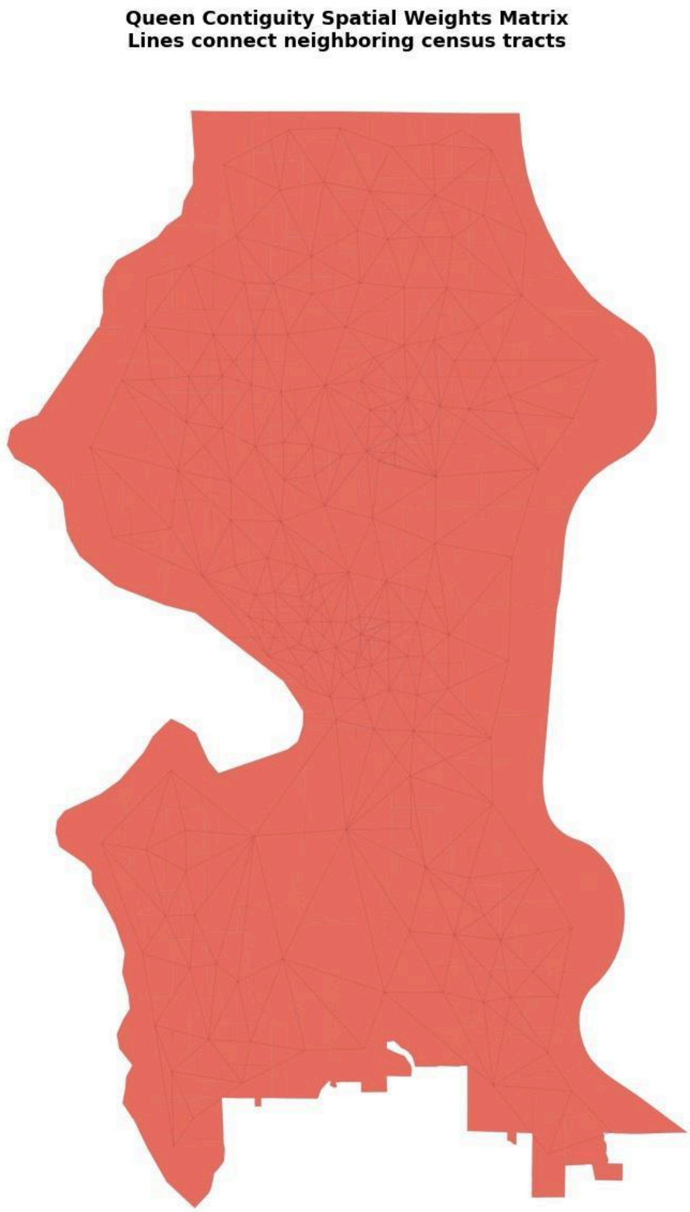
Side-by-Side Small Multiples 6 variables with black outlines on top 5 tracts per variable. Three distinct spatial signatures visible: (1) Crime/Police/EMS Downtown/South corridor; (2) Income inverse North/East pattern; (3) Commercial/Transit North-South urban spine.

Plot Observation: Comparing the six maps simultaneously reveals three distinct spatial signatures. Crime rate, police rate, and EMS rate share an almost identical spatial footprint dark in Downtown/South Seattle, light in the residential periphery visually confirming their r=0.98 and r=0.74 correlations. Median income shows the inverse pattern dark green (high income) in the north and east where crime is low. Commercial ratio and transit access share a third pattern concentrated in the urban core along the north-south spine consistent with their moderate positive correlations with crime (r=0.32, r=0.29). The three spatial clusters visible across these maps foreshadow the predictor groupings that will emerge in the regression analysis in Phase 5.

3.6 Spatial Autocorrelation Analysis

Spatial autocorrelation analysis tests whether the spatial distribution of crime is random or clustered, and identifies specific hotspot and coldspot tracts.

3.6.1 Spatial Weights Matrix

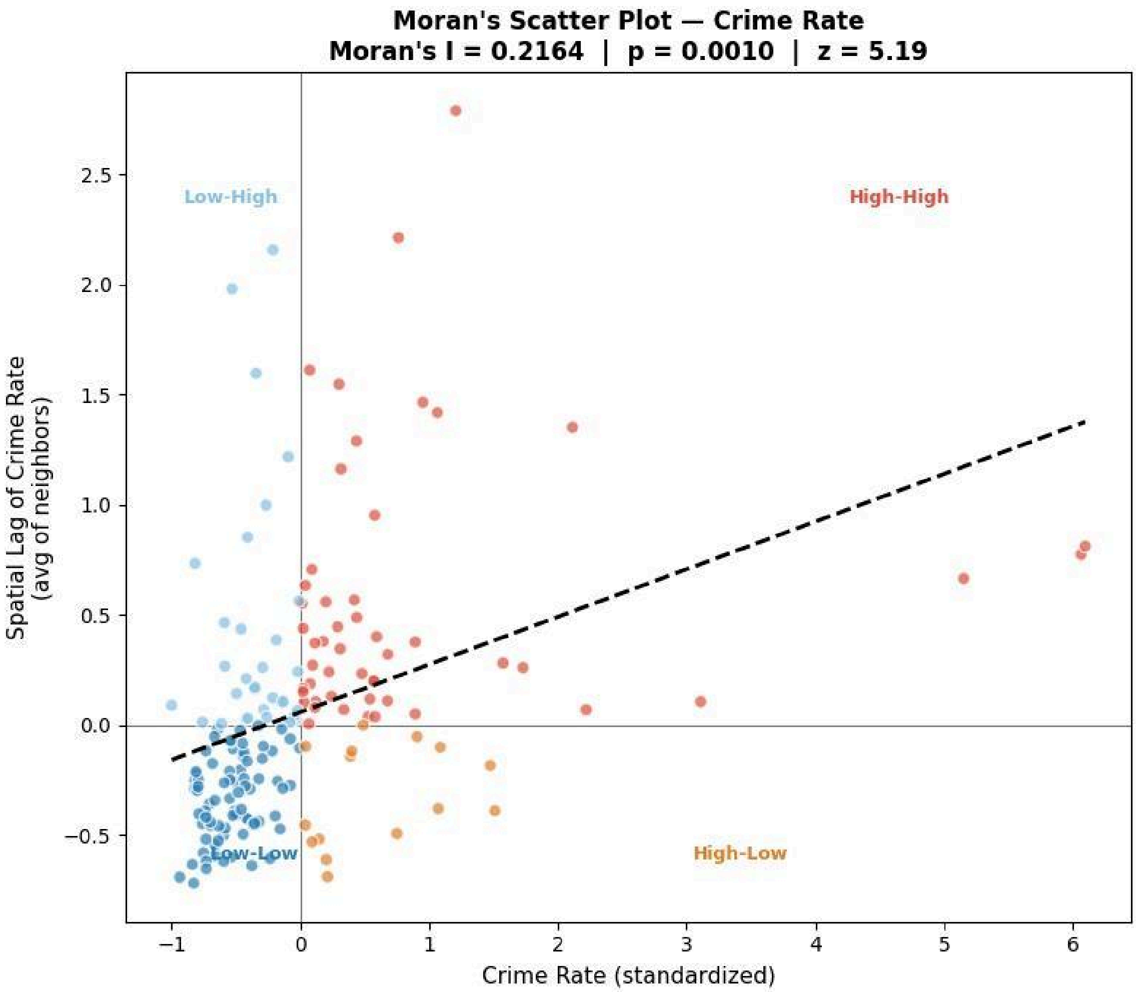


Queen Contiguity Spatial Weights Matrix lines connect neighboring census tracts. Denser triangulation in the urban core where tracts are smaller.

Observation: The Queen contiguity weights matrix was successfully constructed for all 177 Seattle census tracts with zero islands confirming every tract shares at least one boundary with a neighbor. The average tract has 5.73 neighbors ranging from 2 (corner or peninsula tracts) to 11 (small central tracts surrounded by many neighbors). The 3.24% non-zero weight density is expected for a contiguous urban boundary; most tract pairs are non-adjacent. The connectivity map confirms a well-connected spatial network with denser triangulation in the urban core where tracts are smaller. The weights matrix is row-standardized and ready for Moran's I computation.

Queen contiguity weights matrix: 177 spatial units, 0 islands, average 5.73 neighbors per tract (range 2–11), 3.24% non-zero weight density. Row-standardized. All tracts fully connected.

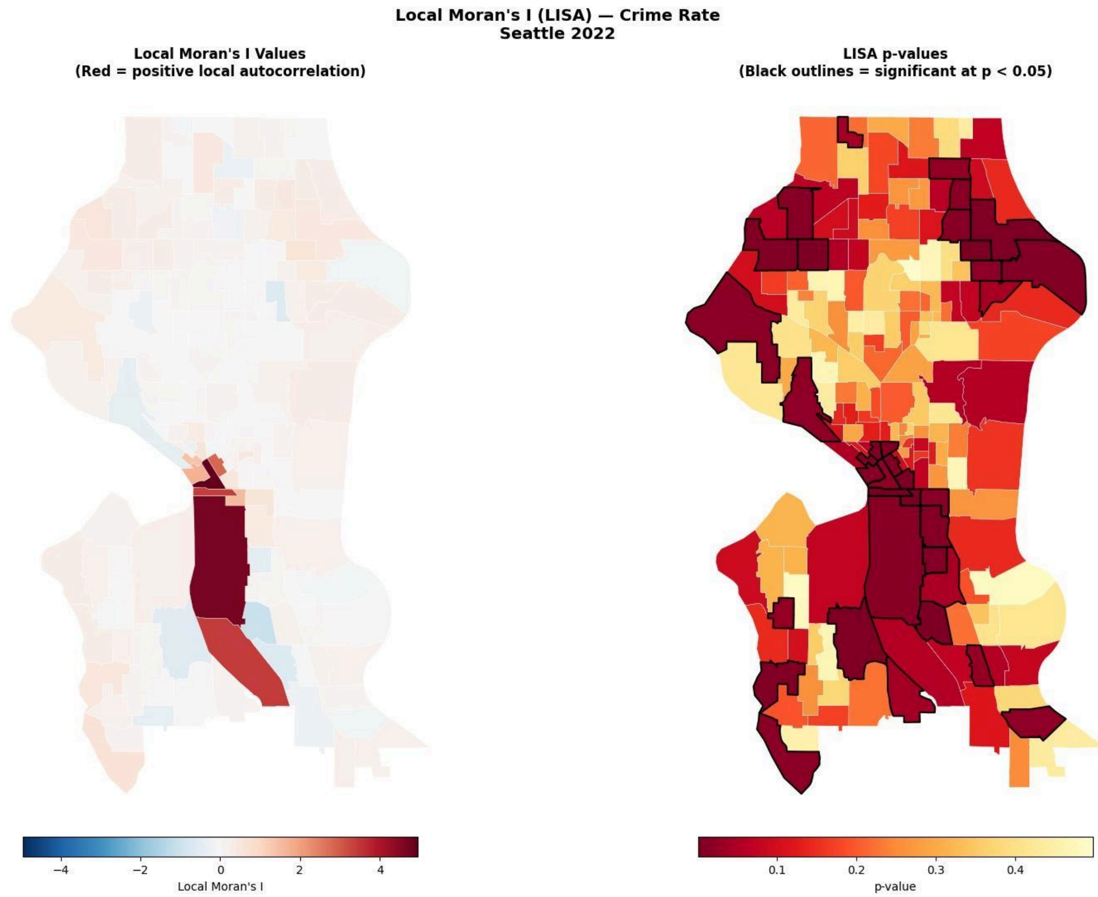
3.6.2 Global Moran's I



Moran's Scatter Plot Crime Rate. Moran's I = 0.2164, p = 0.001, z = 5.19. Points colored by quadrant: High-High (red), Low-Low (blue), High-Low (orange), Low-High (light blue).

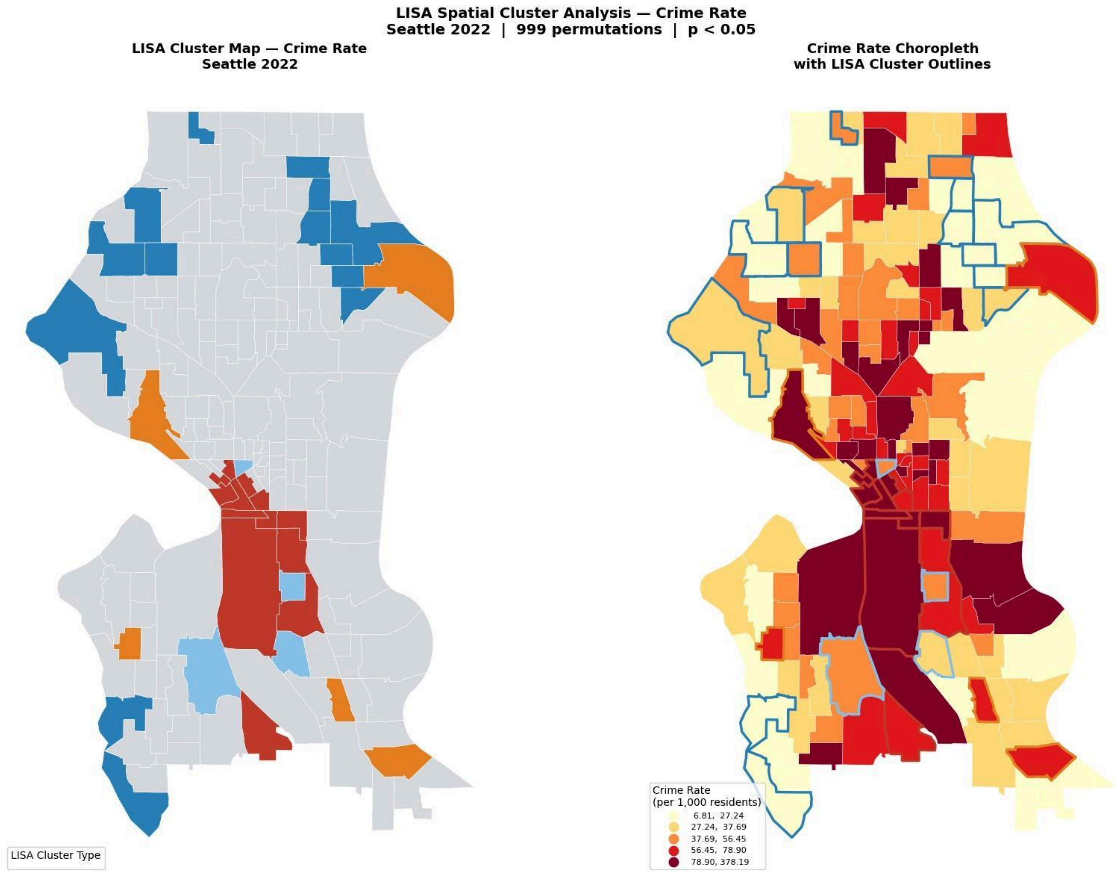
Plot Observation: Global Moran's I = 0.2164 (p = 0.001, z = 5.19, 999 permutations) confirms statistically significant positive spatial autocorrelation. Crime is not randomly distributed across tracts high-crime tracts clustered near other high-crime tracts. This has a direct implication for modeling: OLS regression residuals are likely to be spatially autocorrelated, violating the independence assumption. Spatial regression models (spatial lag or spatial error) must be evaluated in Phase 5.

3.6.3 Local Moran's I (LISA)



Local Moran's I values (left) and p-values with significant tract outlines (right). Extreme positive local autocorrelation (deep red) in Downtown/SoDo. Weak negative local autocorrelation (light blue) at North Seattle periphery.

Plot Observation: The LISA results show 36 statistically significant tracts (20.3% of Seattle), forming two clear spatial clusters: a strong Downtown/South Downtown hotspot with very high positive local autocorrelation, and a North Seattle coldspot with consistently low values. Most central residential tracts show no significant clustering. Of the 36 significant tracts, 13 are High‑High hotspots, 14 are Low‑Low coldspots, and the remaining 9 are spatial outliers (5 High‑Low, 4 Low‑High). The slightly larger number of Low‑Low tracts reflects Seattle's geography, where the extensive low‑crime residential periphery produces more coldspots than the compact but intense Downtown hotspot.



LISA Cluster Map (left) and Crime Rate Choropleth with LISA outlines (right). 36 significant tracts (20.3%): 13 High-High hotspots, 14 Low-Low coldspots, 5 High-Low spatial outliers, 4 Low-High spatial outliers.

Plot observation: The LISA results reveal a clear spatial structure in Seattle's crime pattern, with 79.7% of tracts showing no significant clustering and two dominant clusters standing out. A High-High hotspot of 13 tracts forms a concentrated corridor through Downtown, Capitol Hill, and SoDo, aligning with the highest crime-rate quantiles. A Low-Low coldspot of 14 tracts spans North and West Seattle, reflecting stable, low-crime residential areas. The analysis also identifies 5 High-Low outliers, where isolated high-crime tracts sit within lower-crime surroundings, and 4 Low-High outliers, low-crime pockets embedded in higher-crime zones near Downtown. Together with a significant Global Moran's I of 0.216 (p = 0.001), these patterns indicate meaningful spatial autocorrelation, suggesting that Phase 5 should evaluate spatial lag or spatial error models alongside OLS to avoid violating independence assumptions.

LISA Cluster Type	Count	Pct of All Tracts
High-High (Hotspots)	13	7.3%
Low-Low (Coldspots)	14	7.9%
High-Low (Spatial Outliers)	5	2.8%
Low-High (Spatial Outliers)	4	2.3%
Not Significant	141	79.7%

LISA cluster summary (999 permutations, $p < 0.05$)

The 13 High-High hotspot tracts form a contiguous cluster through Downtown, Capitol Hill, and SoDo confirmed by the right-hand choropleth where their crime rates all fall in the darkest quantile (78–378 per 1,000). The 5 High-Low spatial outliers are the most analytically interesting individual high-crime tracts embedded within lower-crime surroundings, likely representing isolated commercial corridors or transit nodes. The LISA findings will guide spatial regression specification in Phase 5.

4. Plan for Completion

4.1 Remaining Data Processing Tasks

- Create log-transformed versions of all six flagged variables (log1p transformation) for regression modeling.
  crime_rate, population_density, transit_access, unemployment_rate, police_incident_rate, EMS_call_rate
- Compute Variance Inflation Factors (VIF) for all predictor combinations to guide model specification, particularly to decide whether to include both police_incident_rate and EMS_call_rate simultaneously.
- Create spatial lag variables for the spatial regression models using the Queen contiguity weights matrix.
- Standardize all predictor variables (z-scores) for regularized regression (Ridge, Lasso) to ensure comparability of coefficients.

4.2 Analysis and Modeling Plan

Final Report will implement and compare four regression models in increasing order of spatial complexity:

- Model 1 - OLS Baseline: log(crime_rate) ~ all 7 predictors (log-transformed where flagged). Establishes the non-spatial performance benchmark.
- Model 2 - OLS with Variable Selection: Use VIF analysis and stepwise selection to identify the best subset of predictors, resolving the EMS/police multicollinearity issue.
- Model 3 - Spatial Lag Model (SLM): Adds a spatially lagged dependent variable (Wy) to account for spatial dependence in the outcome. Appropriate if the spatial process is substantive.
- Model 4 - Spatial Error Model (SEM): Models spatial dependence in the error term rather than the outcome. Appropriate if spatial autocorrelation reflects omitted spatially structured variables.
- Model 5 (Stretch goal) - Random Forest / Gradient Boosting: Non-parametric models that can capture non-linear relationships and interaction effects. Feature importance analysis to validate predictor rankings from EDA.

4.3 Evaluation and Validation Plan

- Model fit: R-squared, Adjusted R-squared, AIC/BIC for model comparison.
- Spatial diagnostics: Moran's I on OLS residuals if significant, confirms need for spatial regression. Lagrange Multiplier tests (LM-Lag vs LM-Error) to select between SLM and SEM.
- Predictive accuracy: 5-fold cross-validation RMSE on log-scale predictions converted back to original scale.
- Residual analysis: QQ plots, residual vs fitted plots, spatial maps of residuals to check for remaining patterns.

- Interpretation: Coefficient tables with standard errors and p-values; partial regression plots for key predictors; spatial visualization of predicted vs actual crime rates.

4.4 Documentation and Reporting Tasks

- Final Report: Full methodology, results, and discussion incorporating all model outputs and spatial analysis.
- GitHub repository: Clean, documented Jupyter notebooks for each phase uploaded to the project repository.
- Visualizations: All key plots exported at high resolution for inclusion in the final report.
- Limitations section: Address REDACTED coordinate bias, census undercount in high-activity areas, and the residential population denominator issue for Downtown tracts.

4.5 Timeline

MM/DD/YYYY	Task
3/1/2026-3/2/2026	Feature Engineering and Preprocessing for Modeling log-transform all flagged variables, compute VIF, build spatial lag variables, standardize predictors
3/3/2026-3/4/2026	Predictive Modeling Model 1 & 2: OLS baseline and variable selection; residual spatial diagnostics; LM tests
3/5/2026-3/6/2026	Model 3 & 4: Spatial Lag and Spatial Error models using PySAL/spreg; compare AIC/BIC with OLS
3/7/2026-2/8/2026	Conclusion and Discussion Model 5 (stretch): Random Forest; feature importance; cross-validation comparison across all models
3/8/2026	Final Report: Write up full results, discussion, limitations; clean GitHub notebooks; submission

5. References

[1] Anselin, L. (1995). Local Indicators of Spatial Association, LISA. Geographical Analysis, 27(2), 93–115.

[2] Anselin, L. (1988). Spatial Econometrics: Methods and Models. Kluwer Academic Publishers.

[3] Brantingham, P. L., & Brantingham, P. J. (1993). Nodes, paths and edges: Considerations on the complexity of crime and the physical environment. Journal of Environmental Psychology, 13(1), 3–28.

[4] Cohen, L. E., & Felson, M. (1979). Social Change and Crime Rate Trends: A Routine Activity Approach. American Sociological Review, 44(4), 588–608.

[5] Moran, P. A. P. (1950). Notes on Continuous Stochastic Phenomena. Biometrika, 37(1/2), 17–23.

[6] Seattle Police Department (2023). SPD Crime Data: 2008–Present. Seattle Open Data Portal. https://data.seattle.gov/

[7] United States Census Bureau (2023). American Community Survey 5-Year Estimates 2022. https://data.census.gov/

[8] Washington State Office of Financial Management (2023). Small Area Estimates Program (SAEP), 2022 Block-Level Population Estimates.

[9] City of Seattle (2023). Current Land Use / Zoning Detail. Seattle GeoData Portal. https://data-seattlecitygis.opendata.arcgis.com/

[10] PySAL Development Team (2023). PySAL: Python Spatial Analysis Library. https://pysal.org/

[11] Rey, S. J., & Anselin, L. (2010). PySAL: A Python Library of Spatial Analytical Methods. The Review of Regional Studies, 37(1), 5–27.

[12] GeoPandas Development Team (2023). GeoPandas: Python tools for geographic data. https://geopandas.org/

[13] Wilson, J. Q., & Kelling, G. L. (1982). Broken Windows: The Police and Neighborhood Safety. The Atlantic Monthly, 249(3), 29–38.